Cut Sheet

Board Design Summary

Style: scrap-chaotic
Dimensions: 12" (305 mm) × 18" (457 mm) × 1-1/2" (38 mm)
Species: Walnut, Maple, Cherry

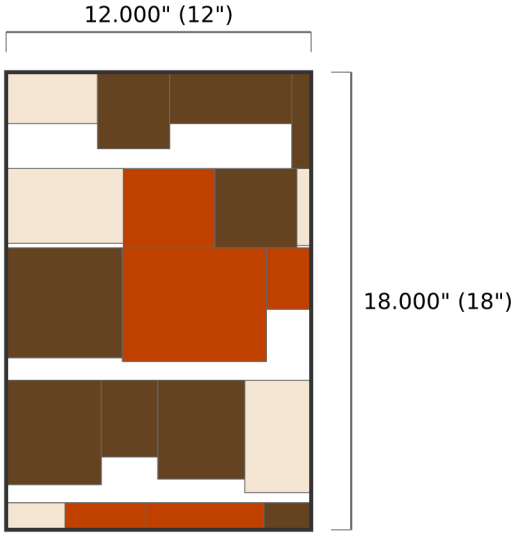
Cut List

Part	Width	Thickness	Length	Species
Board strips	1.000" (1")	1.500" (1 1/2")	18.000" (18")	Walnut

Cut List

Part	Qty	Width	Thickness	Length	Species
Panel A: Maple/Walnut strip-set panel 12" (305 mm) wide: 1 × Maple 3-19/32" (91 mm) (rip 3-21/32" (93 mm), then joint) + 1 × Walnut 2-27/32" (72 mm) (rip 2-29/32" (74 mm), then joint) + 1 × Walnut 4-13/16" (122 mm) (rip 4-7/8" (124 mm), then joint) + 1 × Walnut 3/4" (19.4 mm) (rip 13/16" (21.0 mm), then joint); too wide for the planer at 12" (305 mm): flatten on the router sled or drum sander; then crosscut sled with stop block (never fence) 10 slices × 1-11/16" (43 mm) each; Too wide for the jointer (6" (152 mm)): flatten one face on a planer sled first, then plane the other face parallel	1	12" (305 mm)	1-27/32" (47 mm)	20-3/16" (513 mm)	Maple/Walnut

Board Layout (Top View)



Step 1: Mill rough stock

Joint one face and one edge on the jointer, then plane to final thickness plus 1/8" (3.2 mm). Rest overnight, then plane to final thickness. Take ~1/64" (0.4 mm) passes for the last two passes.

Settings:

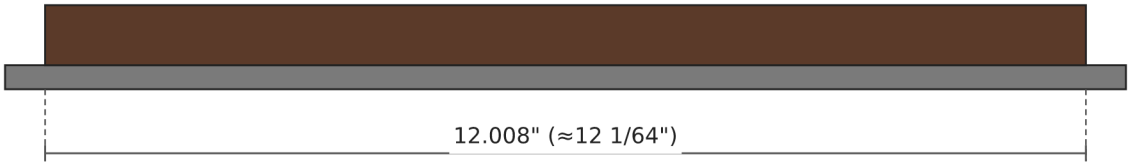
Final pass depth: ~1/64" (0.4 mm)

WARNING: Use push blocks on the jointer. Never joint stock shorter than 12" (305 mm) or thinner than 1/2" (12.7 mm).

WARNING: Keep hands clear of the infeed and outfeed. Feed only boards at least 12" (305 mm) long.

Scrap Chaotic (End Grain): step 1: Mill rough stock

Whole blank on the jointer (schematic, not to scale)



Mill and rip whole blanks before any crosscut. Blank length = max(305 mm, slices x (slice + kerf) + 2 x 25 mm trim). Never send a crosscut slice or piece shorter than 305 mm to the jointer or planer.

Walnut strip: blank 12.008" (≈12 1/64") long, 1 slices of 4.117" (≈4 7/64") crosscut after milling.

Walnut strip: blank 12.008" (≈12 1/64") long, 1 slices of 5.106" (≈5 7/64") crosscut after milling.

Walnut strip: blank 12.008" (≈12 1/64") long, 1 slices of 4.117" (≈4 1/8") crosscut after milling.

Walnut strip: blank 12.008" (≈12 1/64") long, 1 slices of 6.729" (≈6 47/64") crosscut after milling.

Maple strip: blank 12.008" (≈12 1/64") long, 1 slices of 5.036" (≈5 1/32") crosscut after milling.

Maple strip: blank 12.008" (≈12 1/64") long, 1 slices of 5.949" (≈5 61/64") crosscut after milling.

Walnut strip: blank 12.008" (≈12 1/64") long, 1 slices of 5.432" (≈5 7/16") crosscut after milling.

Maple strip: blank 12.008" (≈12 1/64") long, 1 slices of 5.124" (≈5 1/8") crosscut after milling.

Walnut strip: blank 12.008" (≈12 1/64") long, 1 slices of 6.416" (≈6 27/64") crosscut after milling.

Maple strip: blank 12.008" (≈12 1/64") long, 1 slices of 6.561" (≈6 9/16") crosscut after milling.

Maple strip: blank 12.008" (≈12 1/64") long, 1 slices of 4.509" (≈4 33/64") crosscut after milling.

Walnut strip: blank 12.008" (≈12 1/64") long, 1 slices of 6.199" (≈6 13/64") crosscut after milling.

Walnut strip: blank 12.008" (≈12 1/64") long, 1 slices of 5.110" (≈5 7/64") crosscut after milling.

Walnut strip: blank 12.008" (≈12 1/64") long, 1 slices of 5.981" (≈5 63/64") crosscut after milling.

Maple strip: blank 12.008" (≈12 1/64") long, 1 slices of 6.514" (≈6 33/64") crosscut after milling.

Maple strip: blank 12.008" (≈12 1/64") long, 1 slices of 3.160" (≈3 5/32") crosscut after milling.

Maple strip: blank 12.008" (≈12 1/64") long, 1 slices of 3.160" (≈3 5/32") crosscut after milling.

Maple strip: blank 12.008" (≈12 1/64") long, 1 slices of 3.160" (≈3 5/32") crosscut after milling.

Walnut strip: blank 12.008" (≈12 1/64") long, 1 slices of 3.160" (≈3 5/32") crosscut after milling.

Blank lengths: catalog blankRule applied to the app cut list by boardviz; the app export does not apply the rule yet.

Joint one face and one edge on the jointer, then plane to final thickness plus 1/8" (3 mm). Rest overnight, then plane to final thickness. Take ~1/64" (0.4 mm) passes for the last two passes.

Final pass depth: ~1/64" (0.4 mm)

WARNING: Use push blocks on the jointer. Never joint stock shorter than 12" (305 mm) or thinner than 1/2" (12.5 mm).

WARNING: Keep hands clear of the infeed and outfeed. Feed only boards at least 12" (305 mm) long.

Step 2: Sort offcuts

Discard pieces with checks, knots or softwood. Only pieces at least 12" (305 mm) long go through the jointer and planer; never put shorter offcuts through either machine, even on a carrier. Sand or hand-plane short pieces to a common thickness, or glue them edge to edge into the panel first and flatten the panel on the router sled or drum sander once it is at least 12" (305 mm) long. Cut pieces of 6" (152 mm) or less only on a sled with a hold-down.

NOTE: Use closed-grain, non-toxic hardwoods only. Avoid spalted, softwood or oily/sensitising exotics on cutting surfaces.

WARNING: Use push blocks on the jointer. Never joint stock shorter than 12" (305 mm) or thinner than 1/2" (12.7 mm).

WARNING: Do not cut pieces shorter than 6" (152 mm) free-hand on the table saw. Use a sled with a hold-down.

Step 3: Glue up

Glue offcuts into rough panels. Alternate clamps above and below the panel and check for flat with a straightedge.
Flatten Panel A, 12" (305 mm) wide:
too wide for the planer at 12" (305 mm): router sled or drum sander.

Settings:

Clamp spacing: about 8" (203 mm)

NOTE: Use a waterproof PVA glue rated for indirect food contact. Clamp until squeeze-out turns rubbery, then scrape.

Scrap Chaotic (End Grain): glue-up 1, step 3: Glue up



Glue offcuts into rough panels. Alternate clamps above and below the panel and check for flat with a straightedge.

Clamp spacing: about 8" (200 mm)

NOTE: Use a waterproof PVA glue rated for indirect food contact. Clamp until squeeze-out turns rubbery, then scrape.

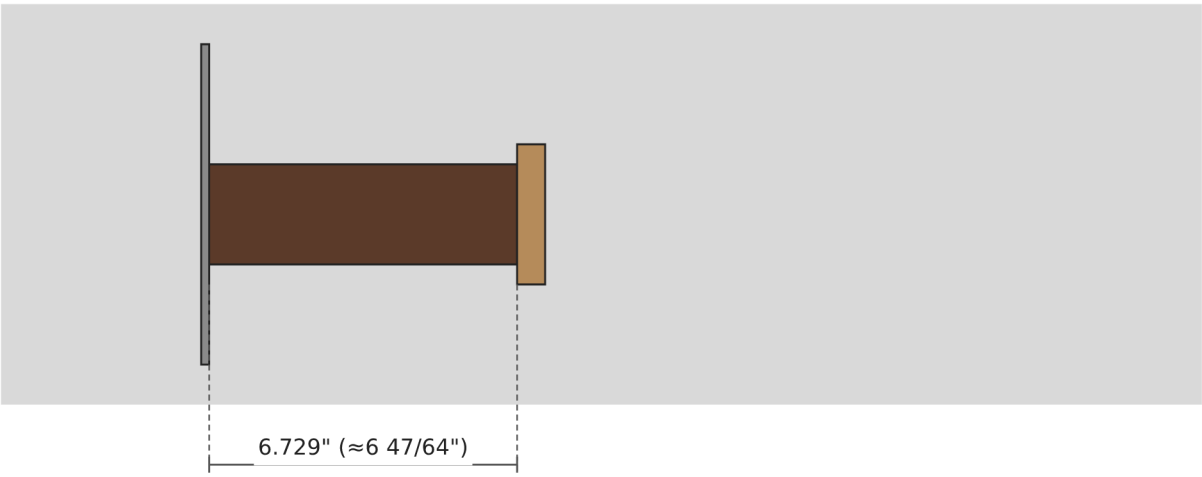
Step 4: Crosscut slices

Crosscut slices on a crosscut sled with a stop block.
Slice size: 1-11/16" (43 mm).
Final thickness: 1-1/2" (38 mm).
End-grain allowance: 3/16" (5.0 mm).
Slices: 10 in all (Panel A 10).
Panel A: sled must clear 12" (305 mm).

DANGER: Never use the rip fence as a length stop for crosscuts. Use a crosscut sled with a stop block set before the blade, so the offcut is never trapped between blade and fence.

WARNING: Do not cut pieces shorter than 6" (152 mm) free-hand on the table saw. Use a sled with a hold-down.

Scrap Chaotic (End Grain): crosscut, step 4: Crosscut slices



Crosscut slices on a crosscut sled with a stop block.

DANGER: Never use the rip fence as a length stop for crosscuts. Use a crosscut sled with a stop block set before the blade, so the offcut is never trapped between blade and fence.

WARNING: Do not cut pieces shorter than 6" (152 mm) free-hand on the table saw. Use a sled with a hold-down.

Step 5: Glue up

Glue slices end grain up in random order. Alternate clamps above and below the panel and check for flat with a straightedge.
Turn every other slice 180° end-for-end; keep the ends flush, with no offset, so the panel gets no longer.

Settings:

Clamp spacing: about 8" (203 mm)

NOTE: Use a waterproof PVA glue rated for indirect food contact. Clamp until squeeze-out turns rubbery, then scrape.

Scrap Chaotic (End Grain): glue-up 2, step 5: Glue up



Glue slices end grain up in random order. Alternate clamps above and below the panel and check for flat with a straightedge.

Clamp spacing: about 8" (200 mm)

NOTE: Use a waterproof PVA glue rated for indirect food contact. Clamp until squeeze-out turns rubbery, then scrape.

Step 6: Flatten and sand

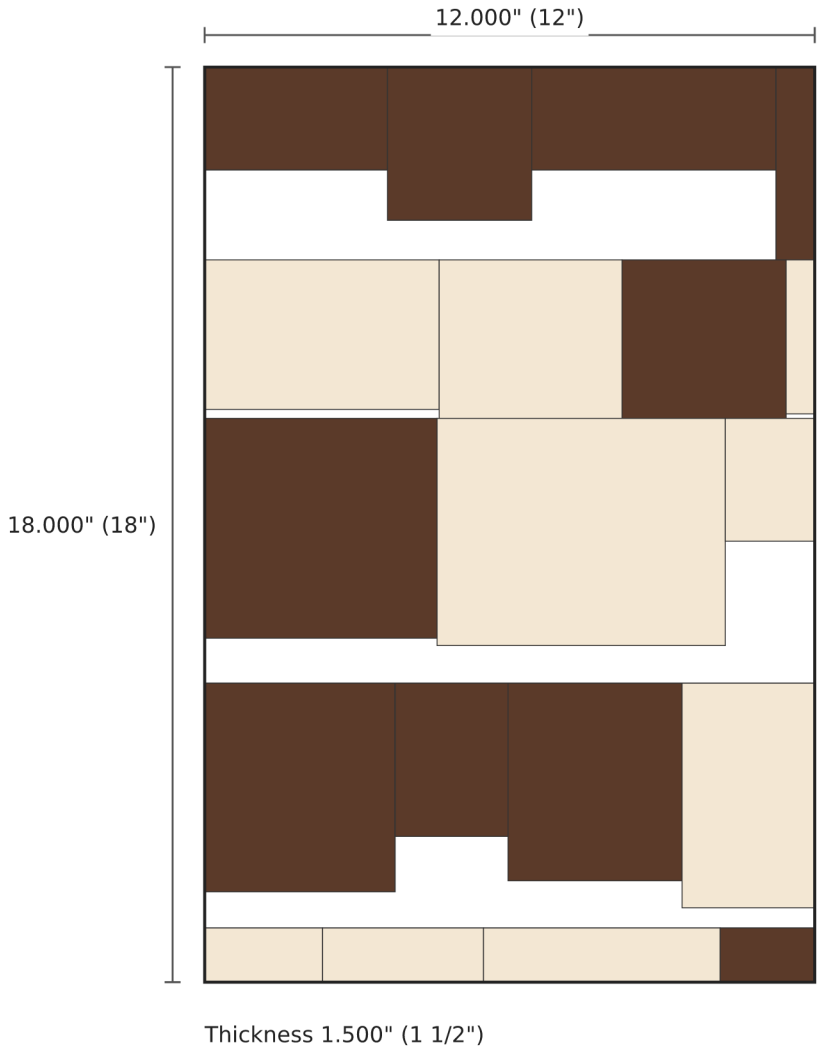
Flatten the final glue-up, 12" (305 mm) wide:
mixed grain: router sled or drum sander, never the planer.
Then sand: 80, 120, 180 grit. Break edges (180 grit).

Settings:

Grits: 80 / 120 / 180

DANGER: Never run an end-grain glue-up through the thickness planer. It can blow apart. Flatten with a router sled or drum sander.

Scrap Chaotic (End Grain): top view



Flatten with a router sled or drum sander, never the thickness planer. Sand 80, 120, 180 grit. Break edges (180 grit).

Grits: 80 / 120 / 180

DANGER: Never run an end-grain glue-up through the thickness planer. It can blow apart. Flatten with a router sled or drum sander.

Step 7: Edge profile

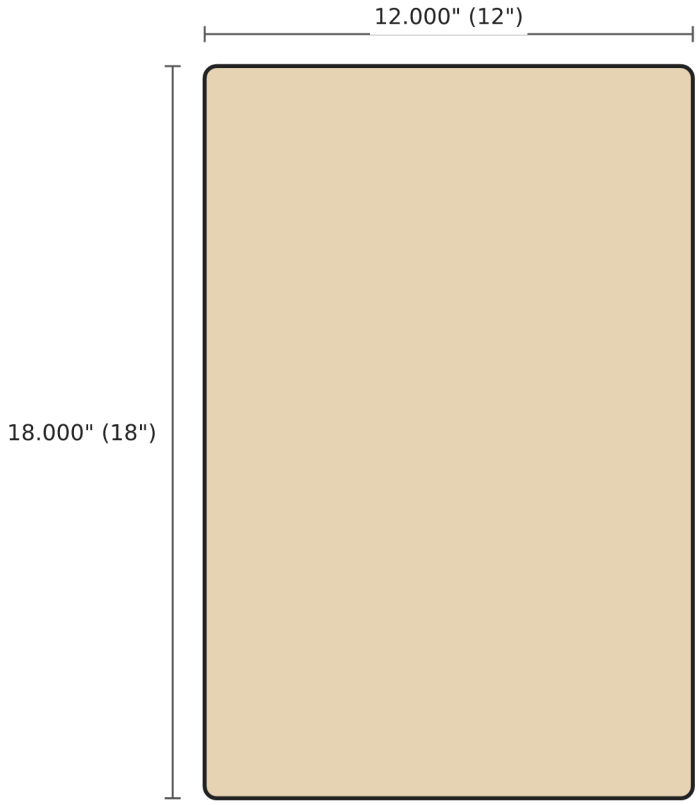
Rout the edge profile, default: 1/8" (3.2 mm) roundover. Optional finger grips: core box / round-nose bit 3/8-1/2" (9.5-12.7 mm).

Settings:

Bit: 1/8" (3.2 mm) roundover

WARNING: Climb cuts can grab. Feed the router against the bit rotation and take shallow passes.

Scrap Chaotic (End Grain): outline, step 7: Edge profile



Rout the edge profile, default: 1/8" (3 mm) roundover. Optional finger grips: core box / round-nose bit 3/8–1/2" (9.5–12.7 mm).

Bit: 1/8" (3 mm) roundover

WARNING: Climb cuts can grab. Feed the router against the bit rotation and take shallow passes.

Step 8: Finish

Raise the grain with a damp cloth, sand 180 again, then flood with food-safe mineral oil and top with board butter.

NOTE: Use closed-grain, non-toxic hardwoods only. Avoid spalted, softwood or oily/sensitising exotics on cutting surfaces.